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Sim-to-Real Performance Analysis of a Custom Quadruped Robot Using Gait Generator-Guided Reinforcement Learning

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Motivation: A Simulated Gait Was Not Enough

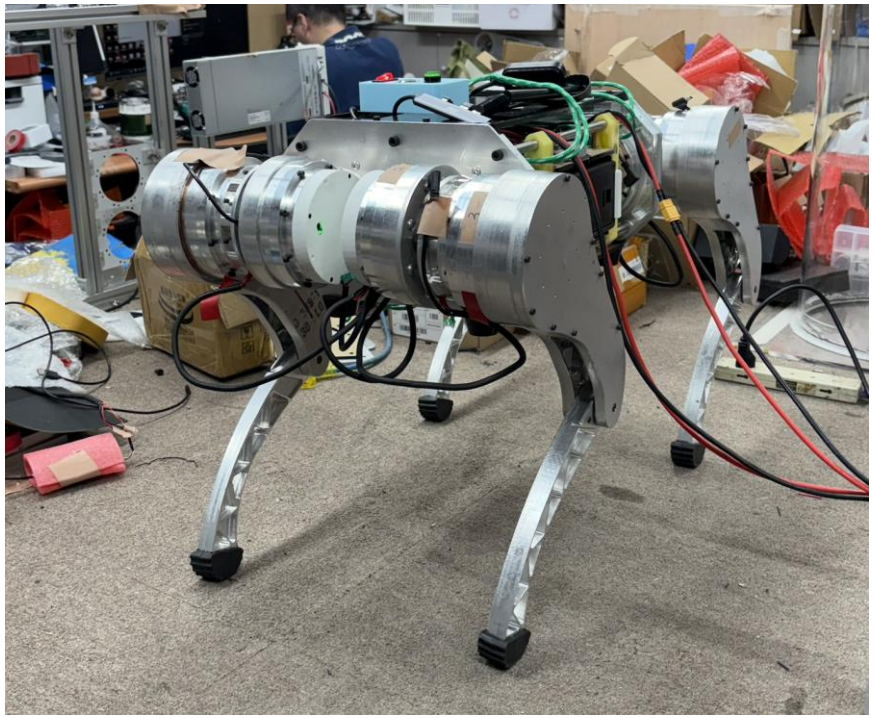
Our lab has mostly used Pure RL, but on this quadruped the hardware swing-foot height was repeatedly lower than expected during sim-to-real transfer.

1. Low Clearance → 2. Toe drag → 3. Imperfect Gait / Hardware Failure

Reward tuning improved some simulations, but it did not reliably remove the hardware foot-clearance failure.

WE WANT TO SOLVE THIS PROBLEM!

Custom Quadruped Build and Modification



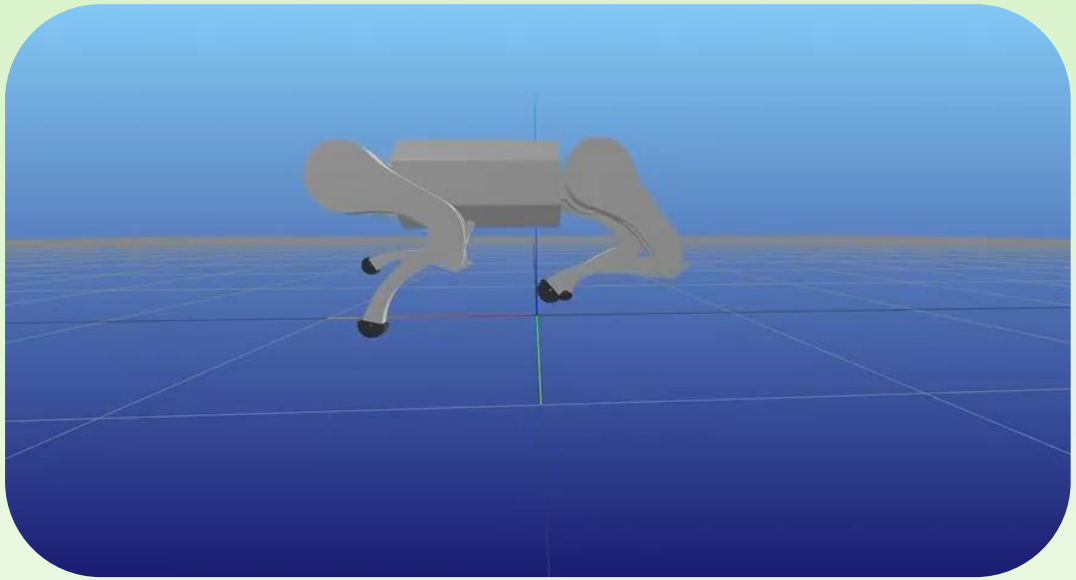
Build Overview

12 DoF: hip / thigh / calf × 4 legs
RO80 motors: 10:1 gearboxes
Link dimensions: 250 mm thigh & calf; 84 mm hip
Custom transmission: friction / backlash / alignment variation
Calf drive: additional 2:1 reduction stage

Modifications Before Tests

Motor placement: lower-resistance motors → heavier-loaded joints
Calf drive: lower gain & stronger shafts
Hip ODrive: firmware & velocity-limit update
Initializing: rear-hip zero offsets & standing-pose match

Pure RL: Simulation vs. Hardware

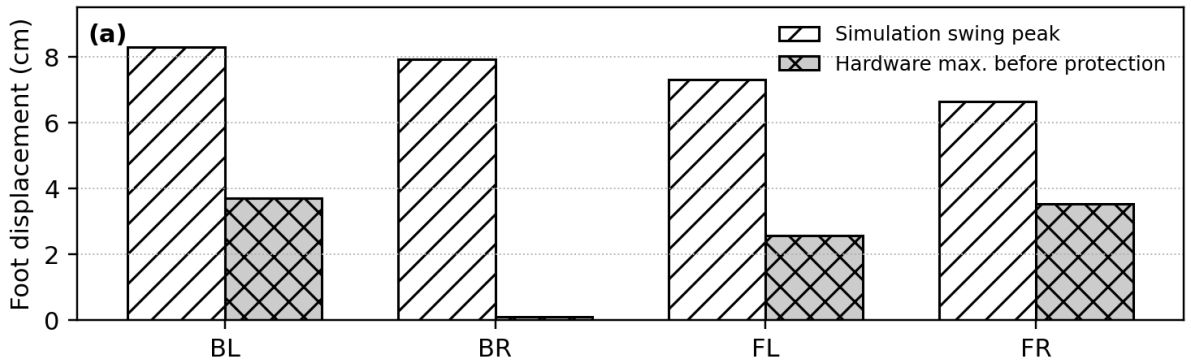


- Stable forward locomotion in simulation
- Smooth gait with sufficient foot clearance
- No obvious failure during walking



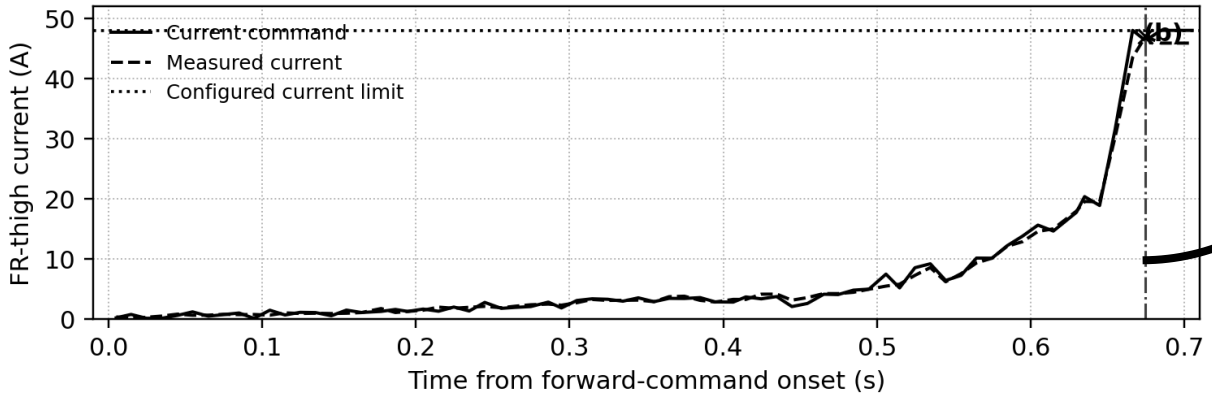
- The robot pitched forward immediately after joystick input
- The front legs entered protection before a full gait cycle was formed
- The simulated policy did not transfer reliably to hardware

Pure RL: Failure Data Analysis



7.54 cm
mean simulated swing peak

2.57 / 3.53 cm
max. FL / FR displacement before protection



0.675 s
time to front-thigh protection

No complete hardware gait cycle was formed

Hidden Risk in Simulation: Insufficient Actuator Margin

70.1% !

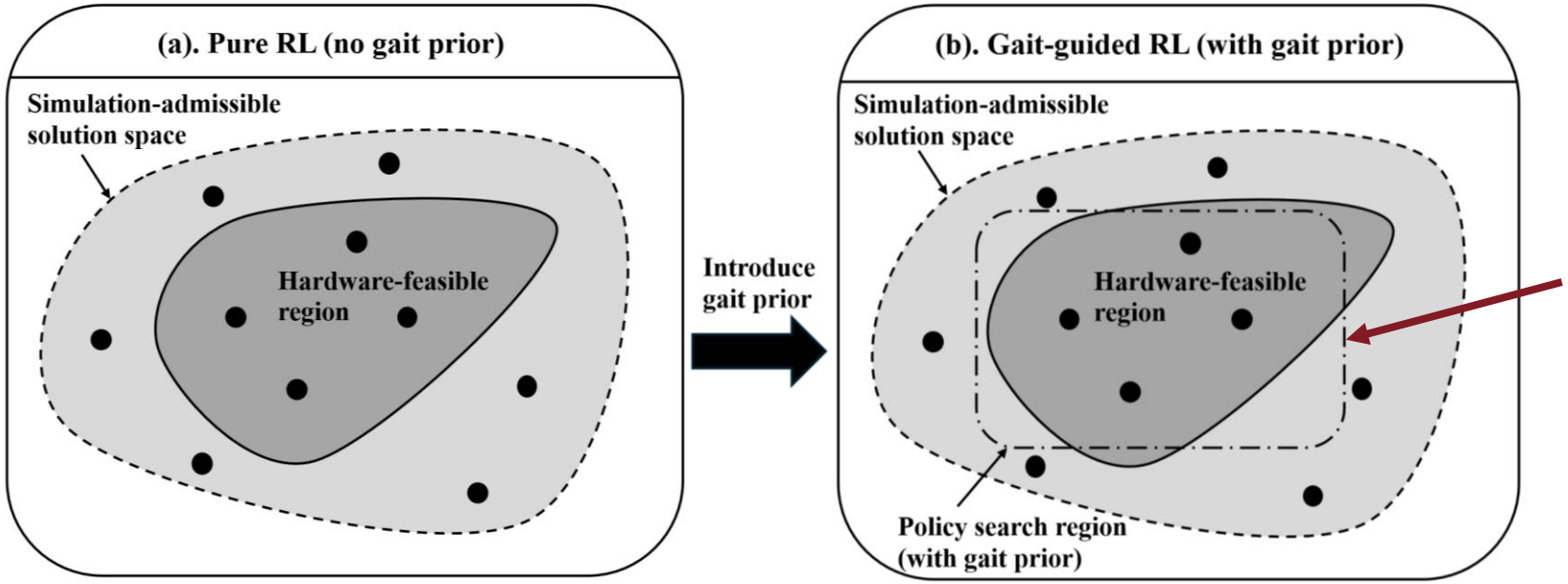
FR-thigh samples at the 28 N·m torque limit

Why this matters

- Simulation can hide policies that are dynamically fragile.
- The gait can look valid while actuator margin is nearly exhausted.
- Small unmodeled contact, friction, or drivetrain losses may push the real robot into protection.

Why Pure RL Failed / Feasibility Mismatch

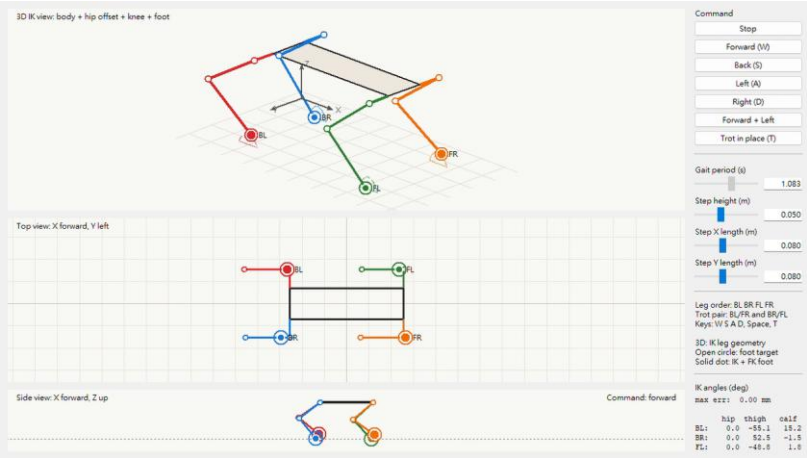
Our Hypothesis:



Why do we expect the gait prior to shift policy search toward this hardware-feasible region?

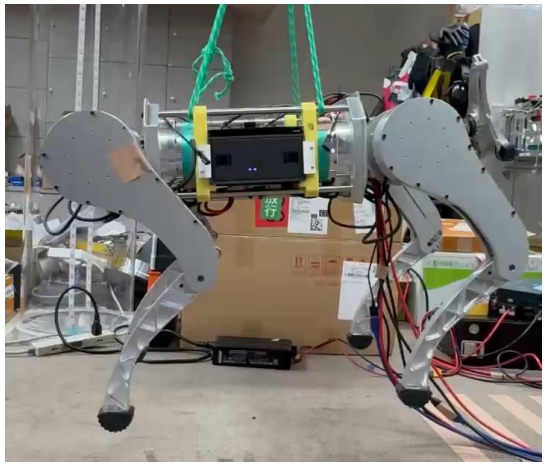
Gait Generator Design + Hardware Verification

Robot-specific gait reference:



- Built from link dimensions and joint conventions
- Inverse kinematics generates joint trajectories
- Tunable gait period, step height, and step length

Direct hardware verification:



(✓ We also tried having it walk on the ground)

- Generated trajectories sent directly to the robot
- No RL policy was used in this check
- Repeated stepping without catastrophic toe drag / protection

First verify feasibility on hardware; then use the reference to guide learning

Gait-Generator-Guided RL

Designing the Gait-Generator-guided RL:

Gait Visualizer & Reference Tuning
- Use our custom gait visualizer
- Tune gait period, step height, and step length
- Preview the generated gait and perform a basic feasibility check

PPO Training + Reference Rewards
- **Retained:** Velocity tracking & stabilization rewards
- **Added:** Joint position / velocity reference tracking
- Foot trajectory reference tracking
- Isaac Lab simulation

Actor + Context Encoder
- Policy inference from robot observations
- Context estimation for adaptation
- Exported as ONNX / TensorRT

Jetson NX + ROS 2 Policy Inference
- Receive joint / IMU observations
- Run Actor + Encoder in real time
- Output joint-position commands

ODrive + RO80 Low-Level Control
- Position control for 12 actuated joints
- Encoder feedback & current monitoring
- Execute commanded joint trajectories

Experimental Setup / Evaluation Metric

Data groups

Pure RL
- 1 simulation playback
- 1 failed hardware record
- No complete gait cycle

Gait-guided RL
- 2 trained controllers
- 3 hardware runs each
- 0.4 m/s straight, level ground

Encoder-consistent clearance metric

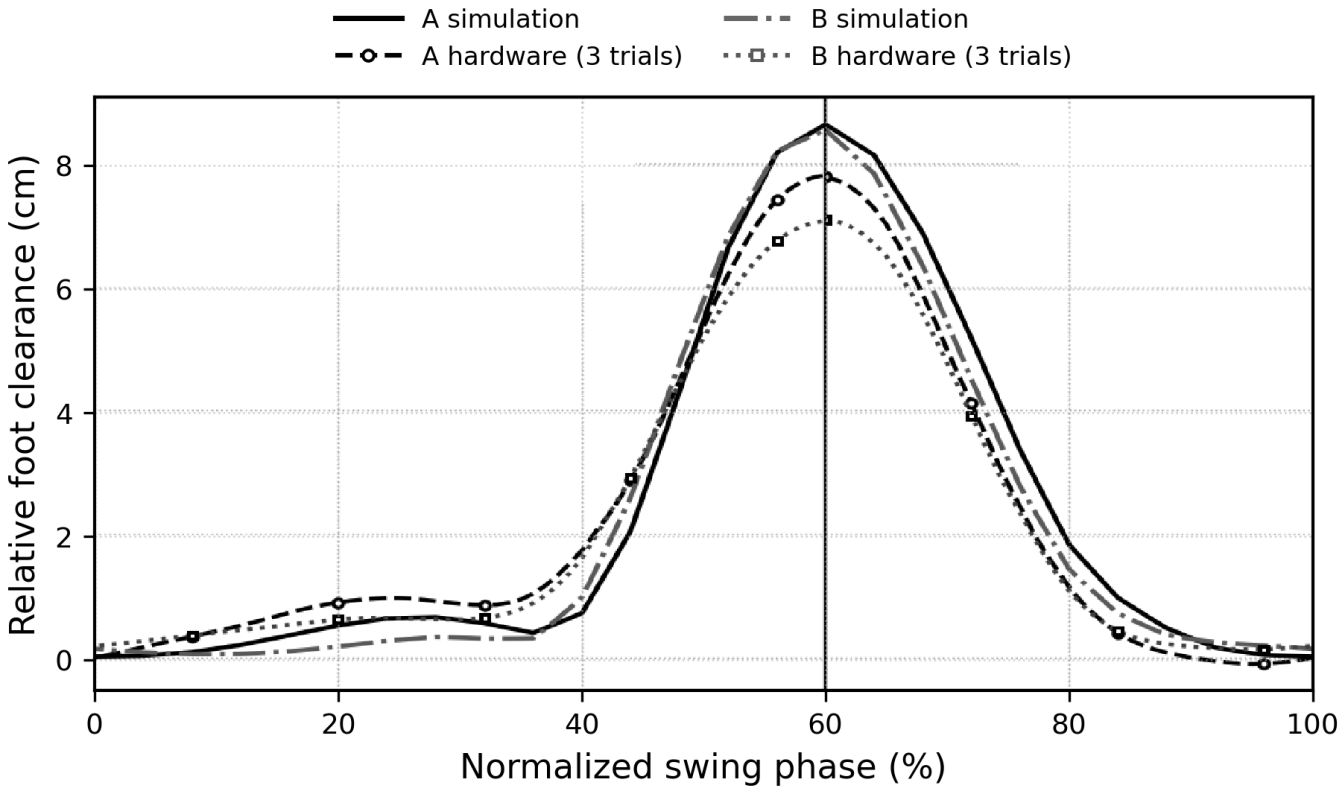
Encoder joint position → Same forward kinematics → 10th percentile stance baseline → Phase-normalized swing cycles

Foot Clearance_i(t) = z_i^B(t) – Q_0.10[z_i^B]

Reported metrics

- **Foot motion:** Mean swing peak, amplitude retention, phase RMSE / correlation
- **Joint tracking:** motor_cmd vs. encoder feedback RMSE
- **Actuator load:** Measured current RMS and peak current
- **Body stability:** Roll / pitch standard deviation when quaternion data are available

Main Result: Foot Clearance Transfer



Guided A

- 8.63 cm sim → 7.78 ± 0.06 cm real
- 90.0% amplitude retention
- Phase correlation: 0.980

Guided B

- 8.54 cm sim → 7.25 ± 0.45 cm real
- 84.9% amplitude retention
- Phase correlation: 0.990

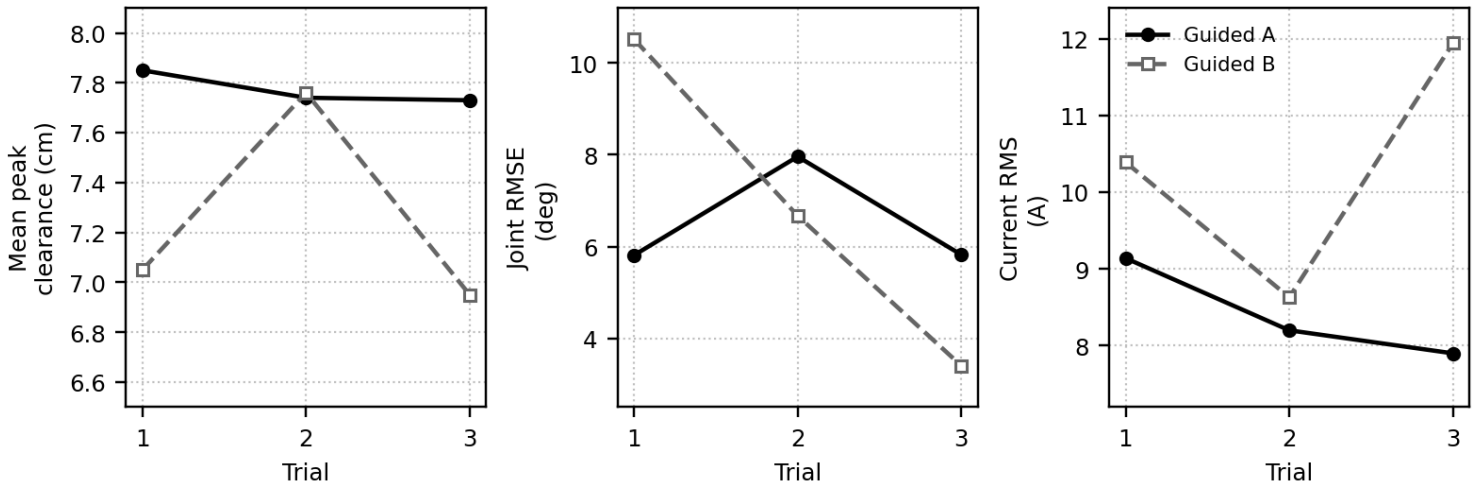
Timing and overall swing shape transfer well; the main residual gap is vertical amplitude loss

Hardware Demonstration: Successful Gait-Guided Locomotion



- 0.4 m/s forward command
- ✓ Sustained periodic swing
- ✓ No front-leg protection during the evaluated run

Repeatability + Practical Implication



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Guided hardware records retained periodic swing and sustained forward motion

Clearance spread

- Guided A: 7.73–7.85 cm
- Guided B: 6.95–7.76 cm

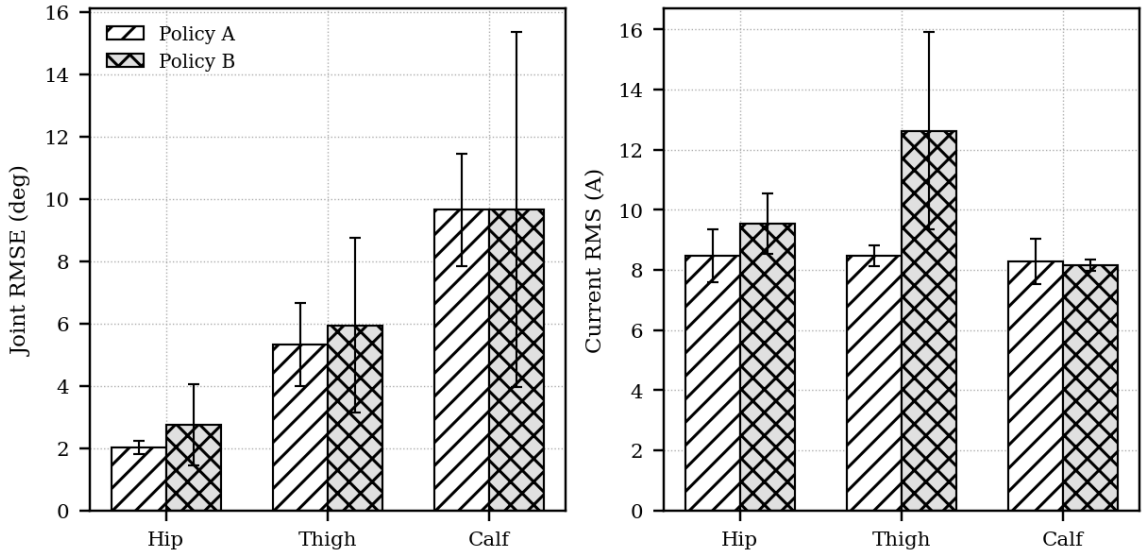
Practical Implication

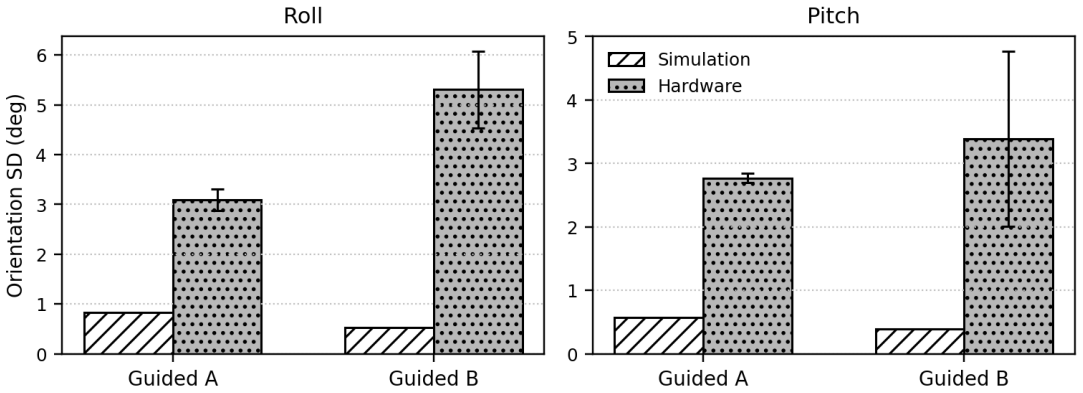
Simulation Performance Can Be Misleading!

Better simulated tracking does not necessarily imply better transferability.

Residual Sim-to-Real Gaps

Gait guidance transfers swing timing, but does not remove drivetrain, disturbance-response, or state-estimation gaps.





Joint execution

Hardware joint RMSE **remains 6.53–6.86°**; **calf ≈ 9.65°** in both guided conditions.

Actuator loading

Current RMS remains moderate, but **brief peaks** can still approach protection range.

Body dynamics

Roll / pitch variation increases strongly on hardware even when foot phase remains consistent.

Conclusion + Future Work

Conclusion

1. Pure RL walked in simulation, but failed before one complete hardware gait cycle.
2. A hardware-tested gait reference narrowed policy search toward executable motions.
3. Guided controllers sustained all six hardware trials and retained 84.9–90.0% swing amplitude.
4. Guided RL improves gait timing, swing shape, and foot clearance, but joint RMSE and body dynamics still need improvement.

Future work

Hardware model
Identify friction, backlash, offsets, transmission efficiency, and effective gains.

State estimation
Improve body velocity, contact, and IMU consistency.

Broader locomotion
Extend beyond 0.4 m/s straight, level-ground walking to turning and uneven terrain.



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Thanks for your attention!